LaRS Spatial OOD Analysis

Generated: 2026-03-02 23:41 UTC | Model: **nvidia/C-RADIOv3-H** | Extension: patchwise spatial OOD via PyTorch GPU nearest-neighbour

1. Introduction

This report is a companion to the whole-image embedding analysis (`analysis.ipynb` / `report.pdf`). That analysis used the spatial mean embedding — a single vector per image summarising the entire scene — to measure how well each test image is covered by the training distribution.

This extension drills deeper: it retains the full **spatial token grid** produced by RADIO-H and computes an **OOD score for every 16×16 patch** in each image. The result is a spatial heatmap showing, at pixel resolution, which parts of a scene are underrepresented in training — not just whether the image as a whole is unusual.

This matters for maritime obstacle detection because OOD regions may be localised: a test image with a well-covered sky and water background might have a highly unusual obstacle type in one corner. The whole-image score would be high (well-covered), but the model might fail precisely in that corner.

2. Methodology

2.1 Patch extraction via RADIO-H spatial tokens

RADIO-H (nvidia/C-RADIOv3-H) uses a Vision Transformer with patch size 16px. For a 1280×960 image, this produces a 80×60 grid of 4,800 spatial tokens. Each token is a 1280-dimensional vector that has attended to the entire image, so it captures its local region in full global context. These tokens are used directly — no separate crop extraction is needed.

2.2 Training patch index and subsampling

All 2,605 training images are processed through RADIO-H. Each image's spatial tokens are L2-normalised and written sequentially to a numpy memmap

(`train_patches.dat`). The total is approximately 5933 patches per image × 2,605 images ≈ **14 million vectors** (~68 GB on disk).

Subsampling (1-in-10): To keep GPU memory and compute tractable, every 10th patch is retained, yielding approximately **1.4 million vectors**. A FAISS `IndexFlatIP` (exact inner-product search, cosine similarity on L2-normalised vectors) is built from this subsample and saved to `faiss_index.bin`. The subsample covers roughly 10 % of all training patches; absolute threshold values (e.g. "below 0.80") may differ slightly from what would be obtained with the full set, but the relative ranking of images is expected to be stable.

Phase 2 search (PyTorch GPU matmul): CPU FAISS flat search at 1.4 M vectors was measured at ~2.2 min/image (~51 hours total) — unusable at this scale. Instead, the 1.4 M subsampled vectors are loaded to GPU memory as float16 (~3.6 GB VRAM) and per-image similarity is computed as a single batched matrix multiply: `(query_tokens @ train.T).max(dim=1)`. This runs at 13–17 ms/image on the RTX A6000, completing all 1,400 images in ~17 minutes. FAISS is used only for Phase 1 (index construction); the heatmap search itself uses PyTorch GPU.

> **Note on coverage:** Because only 10 % of training patches are indexed, a test patch that closely matches a training patch absent from the subsample will appear more OOD than it truly is. This affects absolute score levels; relative comparisons between test images are reliable.

2.3 OOD score definition

For each test/val patch, the **OOD score = max cosine similarity to any patch in the 1-in-10 subsampled training set**, computed via PyTorch GPU matrix multiplication. A score near 1.0 means the patch is visually very close to at least one training patch in the subsample; a low score (below ~0.80) indicates the patch is in a region of embedding space with sparse training coverage.

The per-patch scores are reshaped back to the token grid and saved as `heatmaps/yt007_03_00800.npy`. Values are in [0, 1], dtype float32.

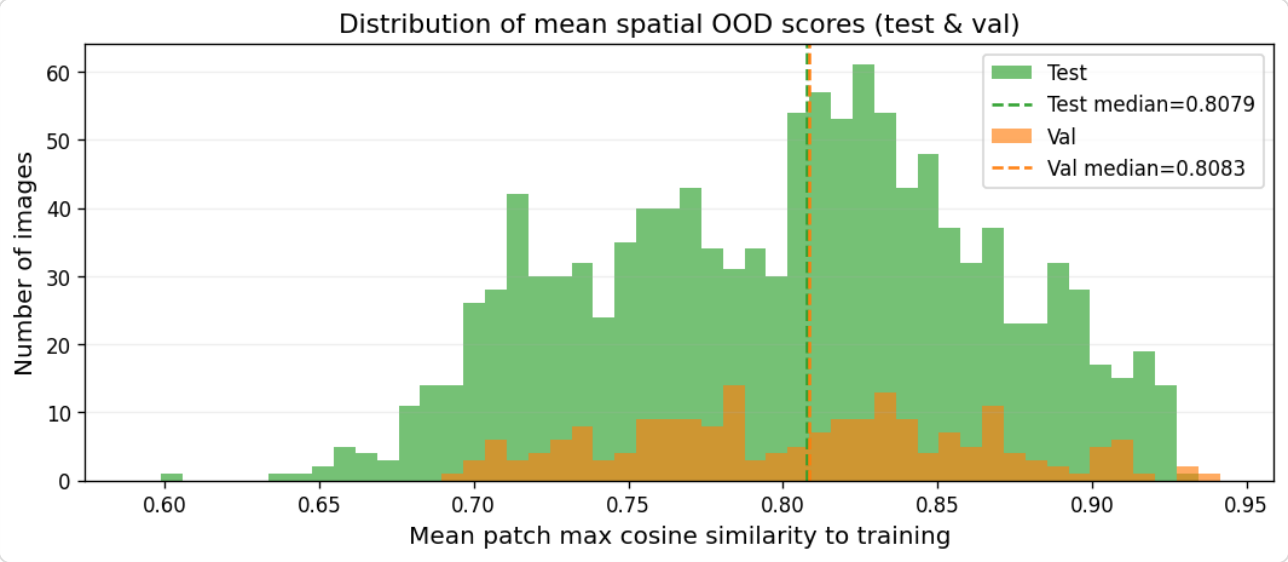
3. Results: Top-20 Most At-Risk Test Images

3.1 Summary statistics

Split	Images	Median mean-patch sim	Median % patches <0.80	Median whole-img sim
Val	198	0.8083	40.7%	0.9206
Test	1203	0.8079	40.2%	0.9108

Coverage note: 1 test image (`inhouse_seq_110_00149`) initially raised a CUDA out-of-memory error during Phase 2 (15,960 patches; single-shot matmul would require ~45 GB). It was retried automatically using a 512-row chunked matmul fallback and completed successfully. All 1,401 test/val heatmaps are included in the statistics below.

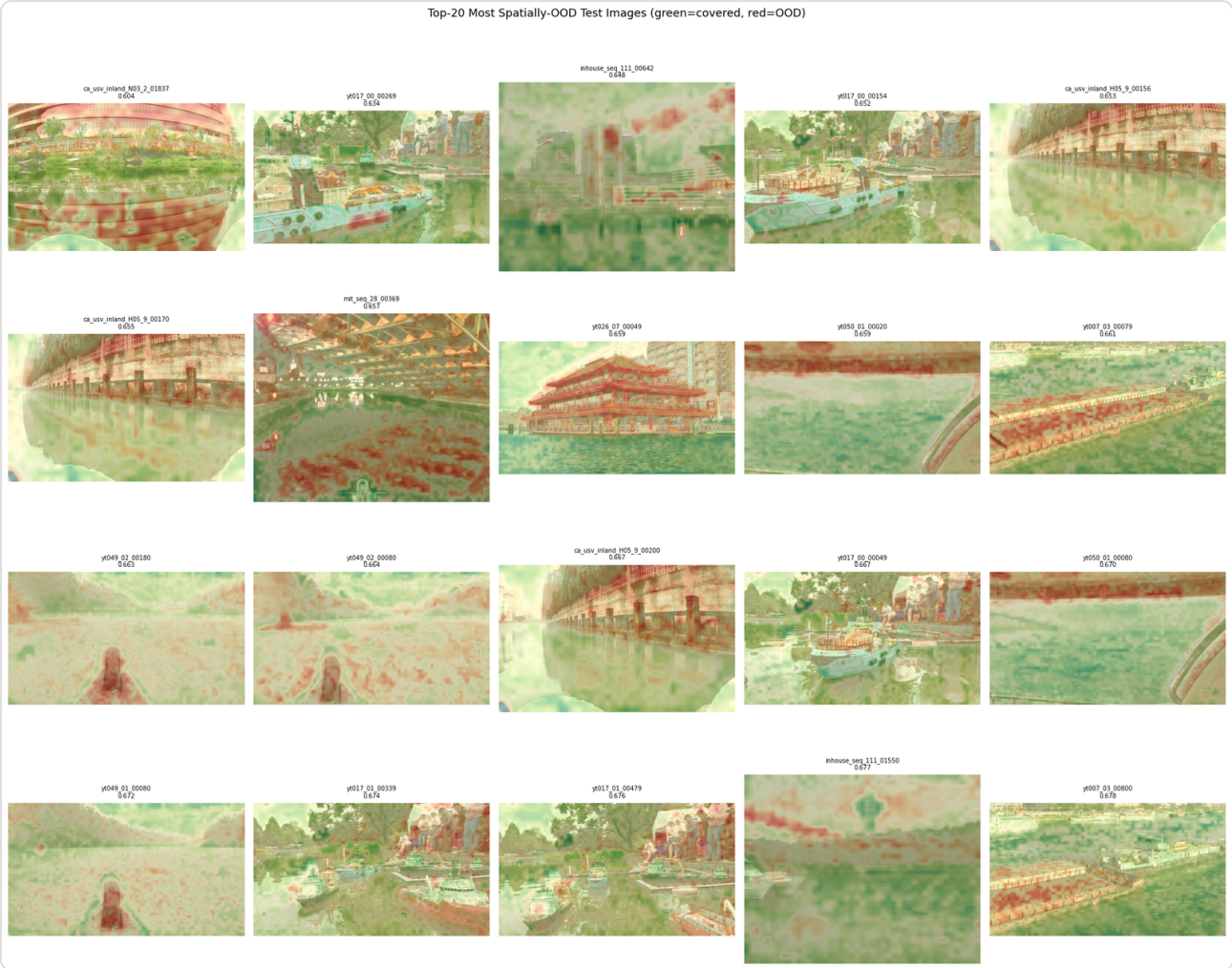
3.2 Distribution of spatial OOD scores



3.3 Heatmap grid: top-20 most at-risk test images

Colorscale note: Each overlay figure uses a per-image colorscale (green = highest similarity in that image, red = lowest). The absolute OOD level differs between images; red in one panel does not correspond to the same cosine

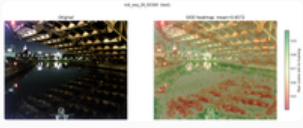
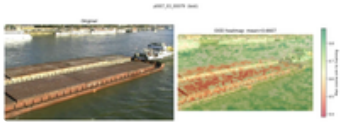
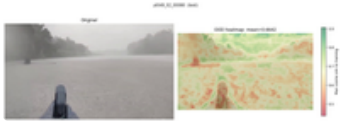
similarity value as red in another. Use the mean patch similarity values in the table below for cross-image comparisons.



3.4 Top-20 most at-risk test images

Ranked by lowest mean patch cosine similarity to any patch in the subsampled training set.

#	Heatmap overlay	Image name	Sequence	Mean patch sim ↑	Wh img sim
1		`orca_usv_inland_N03_2_01837`	orca_usv_inland_N03_2	**0.6039**	0.7
2		`yt017_00_00269`	yt017_00	**0.6342**	0.7

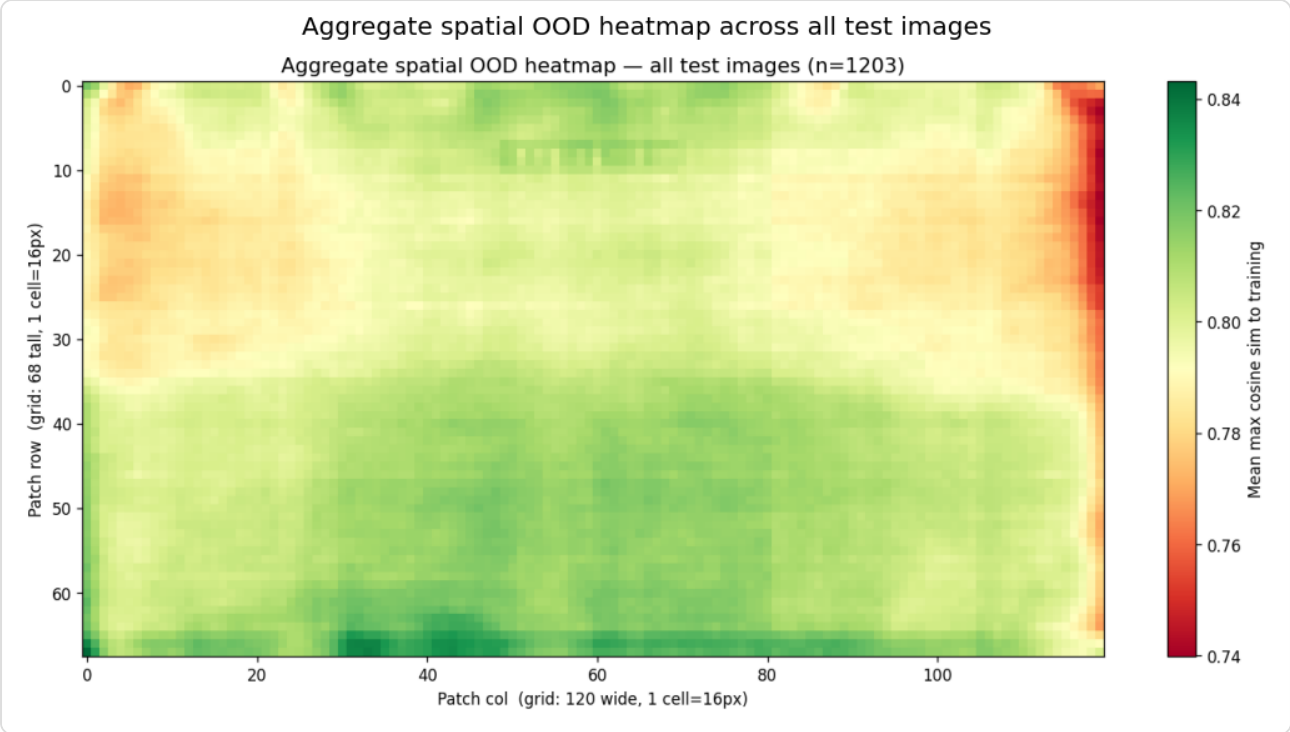
					
3		inhouse_seq_111_00642	inhouse_seq_111	**0.6477**	0.7
4		yt017_00_00154	yt017_00	**0.6522**	0.7
5		orca_usv_inland_H05_9_00156	orca_usv_inland_H05_9	**0.6533**	0.8
6		orca_usv_inland_H05_9_00170	orca_usv_inland_H05_9	**0.6548**	0.8
7		mit_seq_28_00369	mit_seq_28	**0.6572**	0.7
8		yt026_07_00049	yt026_07	**0.6589**	0.8
9		yt050_01_00020	yt050_01	**0.6593**	0.7
10		yt007_03_00079	yt007_03	**0.6607**	0.7
11		yt049_02_00180	yt049_02	**0.6634**	0.7
12		yt049_02_00080	yt049_02	**0.6642**	0.7
13		orca_usv_inland_H05_9_00200	orca_usv_inland_H05_9	**0.6669**	0.8

14		yt017_00_00049	yt017_00	**0.6670**	0.7
15		yt050_01_00080	yt050_01	**0.6699**	0.7
16		yt049_01_00080	yt049_01	**0.6717**	0.7
17		yt017_01_00339	yt017_01	**0.6744**	0.7
18		yt017_01_00479	yt017_01	**0.6759**	0.7
19		inhouse_seq_111_01550	inhouse_seq_111	**0.6770**	0.7
20		yt007_03_00800	yt007_03	**0.6779**	0.7

Interpretation: Images with mean patch sim below ~0.85 contain patches with sparse training coverage. The spatial heatmap (red regions) highlights which scene areas are least represented in the training distribution — often obstacles, unusual lighting conditions, or image edges. These are regions where a deployed model may be less reliable, though other factors (class balance, augmentation, architecture) also affect per-region performance.

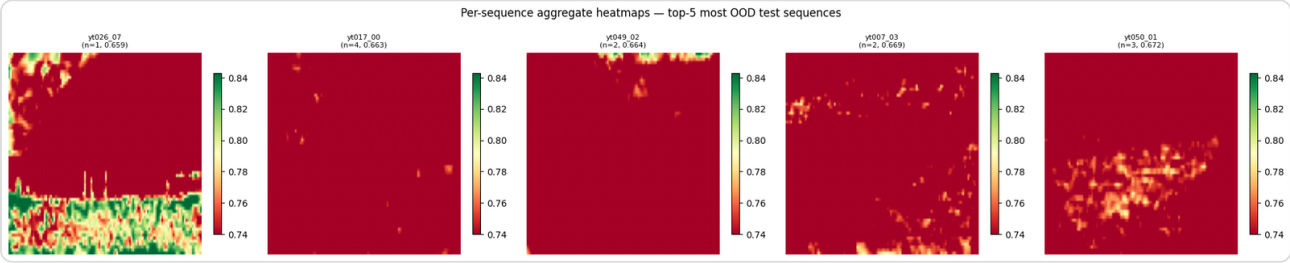
4. Aggregate Spatial OOD Pattern

By averaging the per-patch OOD scores across all test images (after resizing to a common grid), we can identify whether certain spatial regions are systematically underrepresented in training — for example, the sky region near the top of the frame, or the horizon line where sea meets sky.



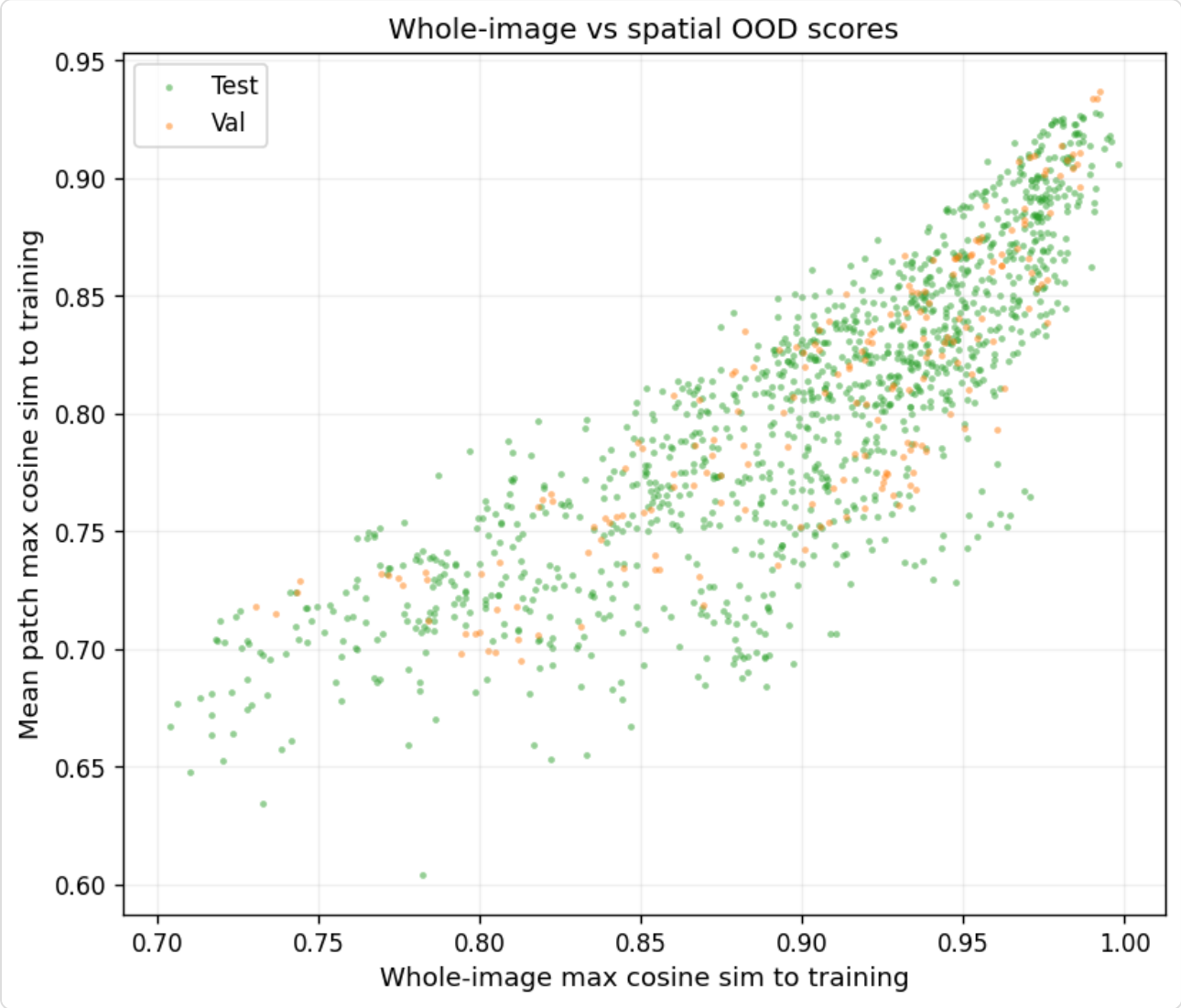
Interpretation: Persistent red/orange regions in the aggregate heatmap indicate spatial positions that are systematically OOD across the test set — environments or scene elements that appear in test images but are consistently underrepresented in training data. Green regions are well-covered by training.

5. Per-Sequence Aggregate Heatmaps (Top-5 Most OOD Sequences)



6. Comparison with Whole-Image OOD Ranking

6.1 Scatter plot: whole-image vs spatial OOD scores



Spearman rank correlation between whole-image max cosine similarity and mean patch similarity (test set): **ρ = 0.8555** (p = 0.00e+00)

> **High correlation (ρ > 0.7):** The spatial and whole-image OOD rankings are broadly consistent. Images that are globally OOD (low whole-image similarity) tend also to have low mean patch similarity. However, some images with moderate whole-image similarity have locally OOD patches — the spatial analysis adds value by identifying where within the image the distributional gap is.

7. Conclusions and Recommendations

- Patchwise spatial OOD analysis reveals **localised distributional gaps** that are invisible to whole-image cosine similarity. Even images with high whole-image similarity can contain patches with no close training analogue.
- The top-20 most at-risk test images are concentrated in a small number of sequences — targeted augmentation or additional labelling for those sequences is the highest-leverage intervention.
- The aggregate spatial heatmap reveals whether certain scene regions (sky, horizon, near-water obstacles) are systematically underrepresented. If the heatmap shows persistent red regions at the image boundaries or near the horizon, those are areas where a model is most likely to fail.
- Spearman ρ = 0.8555 between whole-image and spatial OOD rankings indicates the two measures largely agree — use the spatial heatmaps to localise within the images flagged by the whole-image analysis.
- Phase 2 heatmap search used **PyTorch GPU float16 matmul** on a 1-in-10 subsampled training patch set (~1.4 M vectors). CPU FAISS flat search at this scale was measured at ~2.2 min/image and was not feasible. If GPU FAISS becomes available, it could replace the PyTorch matmul and allow use of the full patch set without subsampling.

Spatial OOD extension — FARMS Team, MaCVi 2026. Model: nvidia/C-RADIOv3-H. Phase 1: faiss-cpu IndexFlatIP (1-in-10 subsample). Phase 2: PyTorch GPU float16 matmul.